

EN 506: Design of Energy Systems

Course Code	EN 506
Course Name	Design of Energy Systems
L-T-P-C	3-0-0-3
Prerequisites	Instructor Consent
Intended for	UG/PG
Mutual Exclusion	None
Approval	9th Senate (May 2016)

Course Contents

- **Module I:** Introduction to energy systems, Introduction to critical components for design in energy system, thermal stresses, material selections (4 Lectures)
- **Module II:** Design of pressure vessels, Stresses in pressure vessels, Autofrettage, Thermal stresses, Design of pressure vessel components such as shell, heads, nozzles, flanges as per ASME & IS codes, Buckling (6 Lectures)
- **Module III:** Design of high pressure power plant piping systems, flow characteristics, material selection, thermal analysis of pipe and joints, thermal insulations (4 Lectures)
- **Module IV:** Design of material handling in power plants, coal conveyor belts, ash handling system, slurry disposal system, handling of radioactive materials (8 Lectures)
- **Module V:** Design of heat transfer equipments, Boilers: classification, selection, Heat exchangers: classification, selection, flow friction and pressure drop analysis, basic thermal design, eNTU, p-NTU, MTD methods. Shell and tube heat exchanger, construction and thermal features, thermal design procedure, Kern method, Bell Delaware method (8 Lectures)
- **Module VI:** Thermal design of regenerators, classifications, design parameters. Design of compact heat exchangers, plate and fin, fin-tube and plate and frame heat exchangers, fouling and corrosion in heat exchangers (6 Lectures)
- **Module VII:** Wind turbine blade design, overview of design criteria and certification guidelines, aerodynamic design, structural design, design and choice of sub-systems and components (6 Lectures)

References

1. J. F. Harvey, **Theory and Design of Pressure Vessels**, CBS Publishers and Distributors, 1987.
2. S. Walas, **Chemical Process Equipment, Selection and Design**, Butterworths Sr. in Chemical Engineering.
3. L. Brownell, E. Young, **Process Equipment Design**, John Wiley and Sons.
4. D. Kern, **Process Heat Transfer**, Tata McGraw-Hill, 2000.
5. Fraas, **Heat Exchanger Design**, 2nd Edition, John Wiley & Sons, 1989.
6. J. Manwell, J. McGowan, A. Rogers, **Wind Energy Explained, Theory, Design and Application**, Wiley, 2012.
7. R. Kulwiec, **Materials Handling Handbook**, ASME, John Wiley and Sons.
8. W. Stoecker, **Design of thermal systems**, 3rd Edition, Tata McGraw-Hill Education.